

Hidden State and Backprop Through Time

Deep Learning

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Contents of This Video

In this video, we will cover:

- The hidden state as memory
- Backpropagation through time (BPTT)
- The vanishing gradient problem
- Why RNNs struggle with long sequences
- Preview of solutions (LSTM)

The Hidden State as Memory

Hidden state h_t encodes:

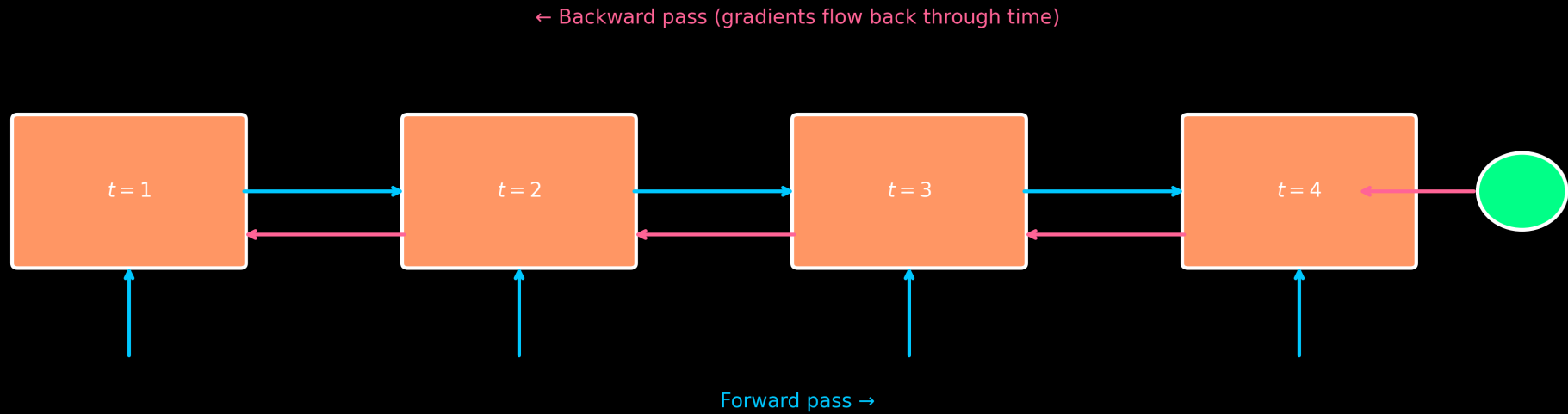
- Information about current input x_t
- Information from all previous inputs $x_1, x_2, \dots, x_{t-1}$

But it's compressed:

- Fixed-size vector (e.g., 128 or 256 dimensions)
- Must somehow encode the entire history
- Older information gets “overwritten”

Backpropagation Through Time (BPTT)

BPTT: Gradients Flow Backward Through All Timesteps



The Vanishing Gradient Problem

Gradient at early timestep depends on many multiplications:

$$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial h_T} \cdot \frac{\partial h_T}{\partial h_{T-1}} \cdot \frac{\partial h_{T-1}}{\partial h_{T-2}} \cdots \frac{\partial h_2}{\partial h_1}$$

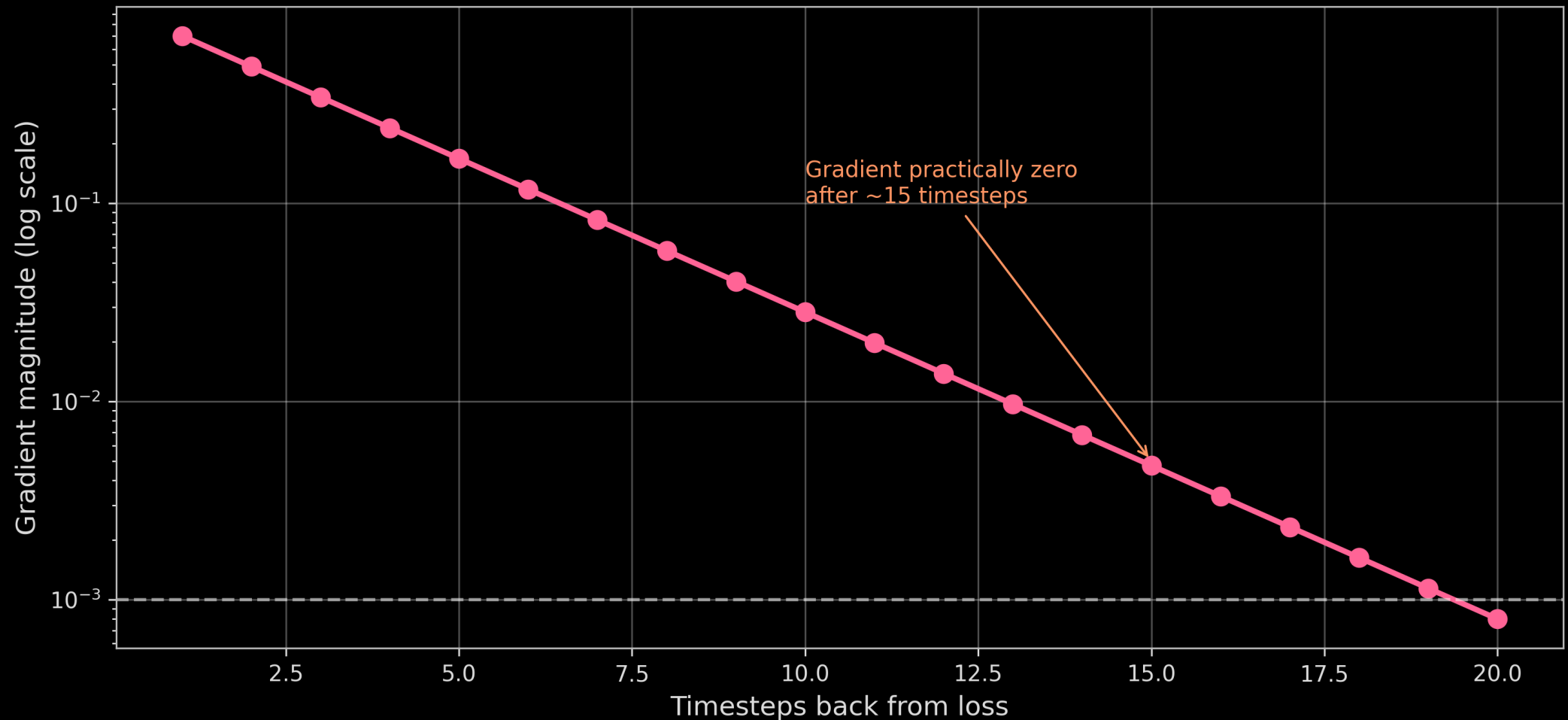
Each $\frac{\partial h_t}{\partial h_{t-1}}$ involves W_{hh} and tanh derivative

If these values are < 1 : gradient shrinks exponentially!

$$\text{Gradient} \propto (\text{small value})^T \rightarrow 0$$

Visualizing Gradient Flow

Vanishing Gradients: Early Timesteps Get No Signal



Exploding Gradients

The opposite can also happen:

- If gradient factors are > 1
- Gradient grows exponentially
- Weights become NaN

Solution: Gradient clipping

$$g \leftarrow \min \left(1, \frac{\text{threshold}}{\|g\|} \right) \cdot g$$

If gradient norm exceeds threshold, scale it down

Consequence: Short-Term Memory

Basic RNNs have short-term memory only:

- Can learn patterns over 10-20 timesteps
- Struggle with long-range dependencies
- Context from early in sequence is lost

Examples of failure:

- “The cat, which sat on the mat, **was** happy” (RNN forgets “cat” is singular)
- “I grew up in France. I speak **French**” (RNN forgets country)

Preview: The Solution

LSTM (Long Short-Term Memory):

- Specialized architecture with “gates”
- Explicitly controls what to remember/forget
- Gradient can flow unchanged through time
- Solves vanishing gradient problem

Coming up in the next video!

What We've Covered

In this video, we've learned:

- Hidden state is a compressed memory of the sequence
- BPTT: gradients flow backward through all timesteps
- Vanishing gradients: many small multiplications $\rightarrow$ gradient shrinks
- Basic RNNs can only learn short-range dependencies (~ 10 -20 steps)
- Gradient clipping prevents exploding gradients
- LSTMs solve the vanishing gradient problem (next video)